Pandas cheatsheet

Pandas is a powerful data analysis and manipulation library for Python. This cheat sheet is a quick reference for Pandas beginners.

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Getting Started

Introduction

You'll need to import pandas to get started:

```
import pandas as pd
```

Creating DataFrames

`pd.DataFrame(data={'col1': [1, 2], 'col2': [3, 4]})`	From a dictionary
`pd.DataFrame(data=[{'a': 1, 'b': 2}, {'a': 3, 'b': 4}])`	From a list of dictionaries
`pd.read_csv('file.csv')`	From a CSV file
`pd.read_excel('file.xlsx')`	From an Excel file

Inspecting Data

`df.head()`	First 5 rows
`df.tail()`	Last 5 rows
`df.shape`	Number of rows and columns
`df.info()`	Info on DataFrame
`df.describe()`	Summary statistics
`df.columns`	Column names
`df.index`	Index
`df.dtypes`	Data types of columns

Selecting Data

`df['col1']`	Select column
`df[['col1', 'col2']]`	Select multiple columns
`df.loc[0]`	Select row by index
`df.loc[:, 'col1']`	Select all rows for 'col1'
`df.iloc[0]`	Select row by position
`df.iloc[0, 1]`	Select specific value
`df[df['col1'] > 2]`	Select rows based on condition

Data Cleaning

`df.dropna()`	Drop rows with any missing values
`df.dropna(axis=1)`	Drop columns with any missing values
`df.fillna(0)`	Replace missing values with 0
`df.drop_duplicates()`	Drop duplicate rows
`df.rename(columns={'old_name': 'new_name'})`	Rename columns
`df.astype('int')`	Change data type

Adding/Removing Data

`df['col3'] = df['col1'] + df['col2']`	Add new column
`df.drop('col1', axis=1)`	Drop column
`df.append(new_row)`	Add new row
`df.insert(2, 'new_col', new_data)`	Insert new column at position 2

Combining Data

`pd.concat([df1, df2])`	Concatenate rows
`pd.concat([df1, df2], axis=1)`	Concatenate columns
`pd.merge(df1, df2, on='key')`	Merge DataFrames on key
`pd.merge(df1, df2, left_on='key1', right_on='key2')`	Merge on different keys
`df1.join(df2, lsuffix='_left', rsuffix='_right')`	Join DataFrames

Aggregating Data

`df['col1'].sum()`	Sum of values in column
`df['col1'].mean()`	Mean of values in column
`df['col1'].count()`	Count of values in column
`df['col1'].min()`	Minimum value in column
`df['col1'].max()`	Maximum value in column
`df['col1'].std()`	Standard deviation
`df['col1'].var()`	Variance
`df.groupby('col1').sum()`	Group by and sum
`df.groupby('col1').mean()`	Group by and mean
`df.groupby(['col1', 'col2']).count()`	Group by multiple columns

Applying Functions

`df.apply(np.sqrt)`	Apply function to all values
`df['col1'].apply(lambda x: x ** 2)`	Apply function to column
`df.applymap(str)`	Apply function to DataFrame elements
`df['col1'].map({'a': 1, 'b': 2})`	Map values
`df['col1'].replace('a', 1)`	Replace values

Handling Dates

`df['date'] = pd.to_datetime(df['date'])`	Convert to datetime
`df['year'] = df['date'].dt.year`	Extract year
`df['month'] = df['date'].dt.month`	Extract month
`df['day'] = df['date'].dt.day`	Extract day
`df.set_index('date', inplace=True)`	Set date as index

Input/Output

`df.to_csv('file.csv')`	Save DataFrame to CSV
`df = pd.read_csv('file.csv')`	Load DataFrame from CSV
`df.to_excel('file.xlsx')`	Save DataFrame to Excel
`df = pd.read_excel('file.xlsx')`	Load DataFrame from Excel
`from sqlalchemy import create_engine`	Import SQLAlchemy for SQL operations
`engine = create_engine('sqlite:///:memory:')`	Create SQL engine
`df.to_sql('table_name', engine)`	Save to SQL table
`df = pd.read_sql('table_name', engine)`	Load from SQL table

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